(2)

Values of Random Variable	0	1	2	3	Total
Number of points in inverse image	10	40	30	4	84

(3)

Values of Random Variable	–20	5	30	Total
Number of points in inverse image	28	48	15	91

(4)

Values of Random Variable	4	5	6	7	8	Total
Number of points in inverse image	1	4	10	12	9	36

EXERCISE 11.2

(1) $f(x)=\begin{cases}\frac{1}{8} & \text{for} \quad x=0,3\\ \frac{3}{8} & \text{for} \quad x=1,2\end{cases}$

(2) (i)

x	2	4	6	8	10	Total
$f(x)$	$\frac{1}{36}$	$\frac{4}{36}$	$\frac{10}{36}$	$\frac{12}{36}$	$\frac{9}{36}$	1

(2) (ii) $F(x)=\begin{cases}0 & \text{for} \quad x<2\\ \frac{1}{36} & \text{for} \quad 2\le x<4\\ \frac{5}{36} & \text{for} \quad 4\le x<6\\ \frac{15}{36} & \text{for} \quad 6\le x<8\\ \frac{27}{36} & \text{for} \quad 8\le x<10\\ 1 & \text{for} \quad 10\le x<\infty\end{cases}$ (iii) $\frac{13}{18}$ (iv) $\frac{31}{36}$

(3) $f(x)=\begin{cases}\frac{1}{4} & \text{for} \quad x=1,3\\ \frac{1}{16} & \text{for} \quad x=0,4\\ \frac{3}{8} & \text{for} \quad x=2\end{cases}$, $F(x)=\begin{cases}0 & \text{for} \quad x<0\\ \frac{1}{16} & \text{for} \quad 0\le x<1\\ \frac{5}{16} & \text{for} \quad 1\le x<2\\ \frac{11}{16} & \text{for} \quad 2\le x<3\\ \frac{15}{16} & \text{for} \quad 3\le x<4\\ 1 & \text{for} \quad 4\le x<\infty\end{cases}$

(4) (i) 8 (ii) $F(x)=\begin{cases}0 & \text{for} \quad x<0\\ \frac{1}{8} & \text{for} \quad 0\le x<1\\ \frac{3}{8} & \text{for} \quad 1\le x<2\\ 1 & \text{for} \quad 2\le x<\infty\end{cases}$ (iii) $\frac{7}{8}$

(5) (i)

x	-1	0	1	2	3
$f(x)$	0.15	0.20	0.25	0.25	0.15

(ii) $P(X<1)=0.35$ (iii) $P(X\ge 2)=0.40$

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